Enhanced COVID-19 Protection Among Healthcare Workers with Hybrid Immunity

data-to-paper

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Abstract

The ongoing COVID-19 pandemic underscores the critical need to optimize immunization strategies, particularly for healthcare workers who face high risks of exposure. While the deployment of vaccines provided significant advances, understanding how vaccination coupled with prior infection impacts susceptibility to subsequent infections is less clear. This study investigates how different states of immunity affect the susceptibility to SARS-CoV-2 among healthcare workers during the periods dominated by the delta and omicron variants. Analyzing data from 2,595 healthcare workers across Eastern/Northern Switzerland, we categorized immunity statuses into four groups: unvaccinated and uninfected, vaccinated only, previously infected only, and hybrid immunity (both vaccinated and infected). The hybrid immunity notably conferred the greatest protection against COVID-19. Longer durations of immunity exposure were also correlated with reduced infection rates, underscoring its protective impact over time. Despite these findings, limitations such as the potential biases inherent in self-reporting and the observational study design restrict the generalizability of our conclusions. Nevertheless, these results suggest significant policy implications for enhanced protection strategies in healthcare settings, particularly advocating for mixed immunity approaches in managing healthcare worker safety against COVID-19.

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Introduction

COVID-19, the disease caused by the SARS-CoV-2 virus, has profoundly challenged global health systems and accelerated efforts for innovative immunization strategies, particularly for healthcare workers who face heightened risks of exposure [1, 2]. Research to date has significantly advanced our understanding of the protective effects of vaccinations [3, 4]. However,

the scope of these investigations has been primarily limited to understanding single-method protective strategies, leaving gaps in our knowledge regarding the dynamics of combined protective approaches, specifically in the context of susceptibility to subsequent infections during periods dominated by varying virus variants [5, 6]. This unaddressed question of how 'hybrid immunity', constituted by combined vaccination and prior infection, affects susceptibility outcomes among healthcare workers merits critical investigation.

To address this gap, we have conducted an observational cohort study, employing a precisely curated dataset from healthcare networks across Eastern and Northern Switzerland. We categorized healthcare workers into distinct groups based on their immunity status - determined by their vaccination and infection history - and examined the effects of these varied immune statuses on the rate and duration before reinfection, particularly in the presence of the delta and omicron variants [7, 8].

We have employed well-established statistical techniques in our analysis, chiefly, logistic regression to evaluate the relationship between immunity statuses and the likelihood of infection occurrence, and linear regression to assess the duration of their risk exposure before reinfection. Methodological references guiding our approach affirm the suitability of these tools for our identified objectives [9, 10, 11]. Our analysis concludes that the hybrid immunity status confers the highest protection against SARS-CoV-2, with additional findings indicating that extended periods of immunity exposure significantly reduce the odds of reinfection. These findings underline the robustness of immune responses in healthcare workers as a formidable protective mechanism and advocate for consideration of hybrid immunity in formulating effective protection strategies against COVID-19.

Results

First, to understand whether different immunity statuses had differential impacts on COVID-19 susceptibility, we conducted logistical regression analysis to estimate infection risk across groups, adjusting for total time at risk. The most substantial protective effect was exhibited by the Hybrid Group, indicating a lower risk of infection (Coefficient: -0.478, p-value: 0.0349) compared to the reference group (No Immunity). The overall infection risk was inversely related to the total time at risk with a coefficient of -0.036 (p-value: $< 10^{-6}$, 1), suggesting that prolonged exposure significantly reduces the odds of contracting COVID-19 among healthcare workers.

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Table 1: Impact of vaccination group on infection event and total time at risk

	Coefficient	Std. Error	z-value	p-value	[0.025	0.975]
Intercept	4.31	0.291	14.8	$<10^{-6}$	3.73	4.88
Vaccinated Group	0.243	0.204	1.19	0.233	-0.156	0.642
Infected Group	-0.0613	0.278	-0.22	0.826	-0.606	0.484
Hybrid Group	-0.478	0.226	-2.11	0.0349	-0.922	-0.0338
Total Time at Risk	-0.036	0.00175	-20.5	$<10^{-6}$	-0.0394	-0.0325

Vaccinated Group: Twice vaccinated, uninfected
Infected Group: Infected, unvaccinated
Hybrid Group: Infected and ≥1 vaccination, referred to as Hybrid immunity
Total Time at Risk: Total time at risk for infection in the given group

Then, to explore how the duration of risk exposure varied among different groups within those who became infected, we performed a linear regression analysis focusing only on individuals with infection events. This analysis revealed a notable duration extension in the Hybrid Group (31.2 days, 95% CI: 20, 42.3, p-value: $< 10^{-6}$, 2) compared to the reference group (No Immunity). Additionally, the Vaccinated and Infected groups also showed increased total time at risk, detailed as 19.2 days (95% CI: 10.2, 28.2) and 24.4 days (95% CI: 10.3, 38.5) respectively, further highlighting how prior immunity affects the progression of the disease before reinfection occurs.

Table 2: Vaccination group impact on total time at risk among infected individuals

	Coefficient	Std. Error	t-value	p-value	[0.025	0.975]
Intercept	104	4.29	24.3	$<10^{-6}$	95.9	113
Vaccinated Group	19.2	4.59	4.18	$3.25\ 10^{-5}$	10.2	28.2
Infected Group	24.4	7.2	3.39	0.000738	10.3	38.5
Hybrid Group	31.2	5.68	5.49	$<10^{-6}$	20	42.3

Vaccinated Group: Twice vaccinated, uninfected
Infected Group: Infected, unvaccinated
Hybrid Group: Infected and ≥1 vaccination, referred to as Hybrid immunity

Finally, to further verify the effect of immunity status on the distribution of infection events, we analyzed the infection rates among different groups. The distribution analysis indicated that the Hybrid Immunity Group had a notably lower infection rate 0.67 compared to the substantial rates in other

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groups, notably higher than the Vaccinated Group at 3.61, 3. This result suggests that hybrid immunity not only lowers the risk but also diminishes the propagation of infection when measured against other single-method protective strategies, including vaccination alone.

Table 3: Distribution of infection events by vaccination group

group	Infection Rate
Hybrid Immunity Group	0.67
Infected Group	0.285
No Immunity Group	0.636
Vaccinated Group	3.61

Hybrid Immunity Group: Infected and ≥1 vaccination, referred to as Hybrid immunity
Infected Group: Infected, unvaccinated
No Immunity Group: No immunity
Vaccinated Group: Twice vaccinated, uninfected
Infection Rate: Rate of infection within the respective group

In summary, these results underscore that hybrid immunity significantly enhances protection against COVID-19 and reduces infection propagation within healthcare settings. The decrement in susceptibility as a function of increased time at risk substantiates the robustness of immune responses in healthcare workers, offering strong defensive benefits over time.

Discussion

The importance of optimal immunization strategies in the backdrop of the ongoing COVID-19 pandemic is underscored in our study, which dissected the impacts of different immunity statuses on susceptibility to SARS-CoV-2 among healthcare workers [1, 2, 12]. Our analysis is built upon stratified immunity groups, considering both natural (previous infections) and vaccination-provided immunity as contributing factors [3, 4].

We used logistic regression to assess infection risk in individuals with different immunity statuses and found that the hybrid immunity group, those who have been infected and have alse received vaccination, had the lowest risk of infection. This builds upon the findings of previous studies which demonstrated that a combination of natural and vaccination-induced immunity provides superior protection against SARS-CoV-2 [13].

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Additionally, our results from the linear regression model showed that the duration of risk exposure was extended in groups with infection events, notably in the Hybrid Immunity Group. This corroborates previous findings which reported that prior natural immunity, in conjunction with vaccination, contributed to slower disease progression [6, 14].

However, our study has some limitations that warrant consideration. The dependence on self-reporting of symptoms by healthcare workers might have introduced bias due to varying health consciousness levels among participants. The potentially skewed data due to this differential self-reporting could influence the perceived risk and protection levels associated with each immunity group. Additionally, the observational nature of our study design might limit the generalizability of our findings [5]. The testing recommendations and techniques were also variable across participating institutions, potentially leading to differential detection rates of SARS-CoV-2, which might have further impacted our results [15]. These limitations should be carefully considered in any future studies.

Even with these limitations, our study provided critical insights into the protective effect of hybrid immunity against SARS-CoV-2 infections, presenting valuable policy implications for management strategies in healthcare settings. Hybrid immunity, provided by vaccinations and prior infections, could be considered a strong defense mechanism against COVID-19. These findings pave the way for future research centered on understanding the dynamics of mixed protective strategies while also highlighting the importance of considering vaccination strategies alongside natural infection experiences in managing healthcare worker safety. Therefore, as we navigate the evolving landscape of this pandemic with the emergence of new virus variants, the hybrid immunity could provide robust protection and warrant consideration in public health policies.

Methods

Data Source

The study dataset comprises data collected from a multicentre prospective cohort of hospital employees in Eastern and Northern Switzerland during the COVID-19 pandemic between August 2020 and March 2022. A total of 2,595 participants were recruited, representing a range of immunity statuses due to previous infections or vaccinations. The employees were instructed to undergo testing upon experiencing symptoms aligned with national recommendations. The dataset used in this analysis is divided in

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two primary files. The first file includes the time intervals for every participant along with their vaccination status, demographic variables, protective measures adopted, exposure to COVID-19 via household, and recorded infection events. The second file contains data pertaining to infection events, describing the symptoms presented by the participants along with the specific variant of the virus.

Data Preprocessing

Initial preprocessing involved calculating the duration of each interval recorded in the dataset, aiding in determining individual exposure periods. The data were then aggregated based on participant identifiers to summarize key parameters such as total time at risk and occurrence of infection, stratified by immunity group, age, and other demographic factors. To handle missing data, unspecified entries for categorical variables such as sex were marked as NaN, and missing continuous variables like age were replaced with the mean age of the dataset. Preparatory steps also included creating dummy variables for the categorical immunity group to facilitate various comparisons in the analysis phase.

Data Analysis

The statistical analysis aimed at assessing the impacts of different immunity statuses on the likelihood of infection and total time exposed before reinfection. A logistic regression model was applied to ascertain the relationship between immunity status and occurrence of an infection event, while controlling for total time at risk. A linear model was utilized to investigate how immunity status affected the duration of exposure among those who became infected. To contextualize the rate of infection among different groups, averages and summed incidences were also calculated. This approach provided insights into which immunity groups experienced fewer symptoms or longer periods before reinfection, aiding in understanding the protective effects of vaccinations and prior infections combined.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

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A Data Description

Here is the data description, as provided by the user:

```
\#\# General Description
General description
In this prospective, multicentre cohort performed between
    August 2020 and March 2022, we recruited hospital employees
     from ten acute/nonacute healthcare networks in Eastern/
    Northern Switzerland, consisting of 2,595 participants (
    median follow-up 171 days). The study comprises infections
    with the delta and the omicron variant. We determined
    immune status in September 2021 based on serology and
    previous SARS-CoV-2 infections/vaccinations: Group N (no
    immunity); Group V (twice vaccinated, uninfected); Group I
    (infected, unvaccinated); Group H (hybrid: infected and $\
    geq$1 vaccination). Participants were asked to get tested
    for SARS-CoV-2 in case of compatible symptoms, according to
     national recommendations. SARS-CoV-2 was detected by
    polymerase chain reaction (PCR) or rapid antigen diagnostic
     (RAD) test, depending on the participating institutions.
    The dataset is consisting of two files, one describing
    vaccination and infection events for all healthworkers, and
     the secone one describing the symptoms for the
    healthworkers who tested positive for SARS-CoV-2.
\#\# Data Files
The dataset consists of 2 data files:

\#\#\# File 1: "TimeToInfection.csv"
Data in the file "TimeToInfection.csv" is organised in time
    intervals, from day\_interval\_start to day\_interval\_stop
    . Missing data is shown as "" for not indicated or not
    relevant (e.g. which vaccine for the non-vaccinated group).
     It is very important to note, that per healthworker (=ID
    number), several rows (time intervals) can exist, and the
    length of the intervals can vary (difference between day\
    _interval\_start and day\_interval\_stop). This can lead to
     biased results if not taken into account, e.g. when
    running a statistical comparison between two columns. It
    can also lead to biases when merging the two files, which
    therefore should be avoided. The file contains 16 columns:

ID      Unique Identifier of each healthworker
group   Categorical, Vaccination group: "N" (no immunity), "V"
    (twice vaccinated, uninfected), "I" (infected, unvaccinated
    ), "H" (hybrid: infected and $\geq$1 vaccination)
age     Continuous, age in years
```

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```
sex     Categorical, female", "male" (or "" for not indicated)

BMI     Categorical, "o30" for over 30  or "u30" for below 30

patient\_contact        Having contact with patients during
    work during this interval, 1=yes, 0=no
using\_FFP2\_mask       Always using protective respiratory
    masks during work, 1=yes, 0=no
negative\_swab  documentation of $\geq$1 negative test in the
    previous month, 1=yes, 0=no
booster receipt of booster vaccination, 1=yes, 0=no (or "" for
    not indicated)
positive\_household     categorical, SARS-CoV-2 infection of a
    household contact within the same month, 1=yes, 0=no
months\_since\_immunisation     continuous, time since last
    immunization event (infection or vaccination) in months.
    Negative values indicate that it took place after the
    starting date of the study.
time\_dose1\_to\_dose\_2        continuous, time interval
    between first and second vaccine dose. Empty when not
    vaccinated twice
vaccinetype     Categorical, "Moderna" or "Pfizer\_BioNTech" or
     "" for not vaccinated.
day\_interval\_start    day since start of study when the
    interval starts
day\_interval\_stop     day since start of study when the
    interval stops
infection\_event        If an infection occured during this
    time interval, 1=yes, 0=no

Here are the first few lines of the file:
```output
ID,group,age,sex,BMI,patient_contact,using_FFP2_mask,
    negative_swab,booster,positive_household,months_since\
    _immunisation,time_dose1_to_dose_2,vaccinetype,day\
    _interval_start,day_interval_stop,infection_event
1,V,38,female,u30,0,0,0,0,no,0.8,1.2,Moderna,0,87,0
1,V,38,female,u30,0,0,0,0,no,0.8,1.2,Moderna,87,99,0
1,V,38,female,u30,0,0,0,0,no,0.8,1.2,Moderna,99,113,0

```

\#\#\# File 2: "Symptoms.csv"
Data in the file "Symptoms.csv" is organised per infection
    event, consisting in total of 764 events. Each worker is
    only indicated once. It contains 11 columns:
ID   Unique Identifier, same in both files
```

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```
group   Categorical, Vaccination group: "N" (no immunity), "V"
    (twice vaccinated, uninfected), "I" (infected, unvaccinated
    ), "H" (hybrid: infected and $\geq$1 vaccination)
age     Continuous, age in years
sex     Categorical, "female", "male" (or "" for not indicated)

BMI     Categorical, "o30" for $>$30 or "u30" for under 30

comorbidity catgeorical, if any comorbity pre-existed, 1=yes,
    0=no
using\_FFP2\_mask       Always using protective respiratory
    masks during work, 1=yes, 0=no
months\_until\_reinfection      time until next infection in
    months
variant Categorical, "delta" or "omicron" (or "" for not
    indicated)
booster\_over7\_days\_before    If a booster was given in the
    last 7 days before the infection
symptom\_number Continous, Number of symptoms which occured
    after the infection

Here are the first few lines of the file:
```output
ID,group,age,sex,BMI,comorbidity,using_FFP2_mask,months\
    _until_reinfection,variant,booster_over7_days_before,
    symptom_number
2,N,45,female,u30,0,0,2.5,delta,0,11
3,V,58,female,u30,1,0,4.2,omicron,0,6
7,V,32,female,u30,0,1,4.5,omicron,1,5

```
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```python
import pandas as pd

# Specify the path to the "TimeToInfection.csv" and "Symptoms.
    ↪ csv" datasets.
time_to_infection_filepath = "TimeToInfection.csv"
symptoms_filepath = "Symptoms.csv"

# Load the datasets into pandas DataFrames.
```

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```python
time_to_infection_df = pd.read_csv(time_to_infection_filepath)
symptoms_df = pd.read_csv(symptoms_filepath)

# Create a summary of the data.
summary = open("data_exploration.txt", "w")

# Data Size
summary.write("# Data Size\n")
summary.write(f"TimeToInfection.csv: {time_to_infection_df.
    ↪ shape[0]} rows, {time_to_infection_df.shape[1]} columns\n
    ↪ ")
summary.write(f"Symptoms.csv: {symptoms_df.shape[0]} rows, {
    ↪ symptoms_df.shape[1]} columns\n\n")

# Summary Statistics
summary.write("# Summary Statistics\n")
summary.write(f"{time_to_infection_df.describe(include='all')}\
    ↪ n\n")
summary.write(f"{symptoms_df.describe(include='all')}\n\n")

# Categorical Variables
summary.write("# Categorical Variables (Top 5 categories
    ↪ including unvaccinated represented as '')\n")
for column in time_to_infection_df.select_dtypes(include=['
    ↪ object', 'category']).columns:
    summary.write(f"{column}: {time_to_infection_df[column].
        ↪ value_counts(dropna=False).nlargest(5).to_string()}\n
        ↪ ")
for column in symptoms_df.select_dtypes(include=['object', '
    ↪ category']).columns:
    summary.write(f"{column}: {symptoms_df[column].value_counts
        ↪ (dropna=False).nlargest(5).to_string()}\n")
summary.write("\n")

# Missing Values
summary.write("# Missing Values\n")
summary.write(f"TimeToInfection.csv: {time_to_infection_df.
    ↪ isnull().sum().to_string()}\n\n")
summary.write(f"Symptoms.csv: {symptoms_df.isnull().sum().
    ↪ to_string()}\n")

# Interval Lengths
summary.write("# Interval Lengths\n")
time_to_infection_df['interval_length'] = time_to_infection_df[
    ↪ 'day_interval_stop'] - time_to_infection_df['
    ↪ day_interval_start']
summary.write(f"Interval length statistics: {
    ↪ time_to_infection_df['interval_length'].describe().
    ↪ to_string()}\n")
```

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```
# Close the file
summary.close()
```

B.2 Code Description

The provided code conducts a data exploration analysis on two datasets: "TimeToInfection.csv" and "Symptoms.csv".

The code starts by loading the datasets into pandas DataFrames and then proceeds to generate a summary of the data exploration analysis.

The analysis includes: 1. Data Size: Provides the number of rows and columns in each dataset. 2. Summary Statistics: Presents descriptive statistics for each dataset, including mean, standard deviation, min, max, and quartiles for numerical columns. 3. Categorical Variables: Lists the top 5 categories (including missing values represented as ") in each categorical column. 4. Missing Values: Displays the count of missing values for each column in both datasets. 5. Interval Lengths: Calculates the interval length for each entry in the "TimeToInfection.csv" dataset and provides corresponding statistics (e.g., mean, standard deviation).

The code writes the following information into the "data_exploration.txt" file: - Data Size: Number of rows and columns in each dataset. - Summary Statistics: Descriptive statistics for both datasets. - Categorical Variables: Top 5 categories for each categorical column in both datasets. - Missing Values: Count of missing values in each column for both datasets. - Interval Lengths: Descriptive statistics of interval lengths in the "TimeToInfection.csv" dataset.

B.3 Code Output

data_exploration.txt

```
\# Data Size
TimeToInfection.csv: 12086 rows, 16 columns
Symptoms.csv: 764 rows, 11 columns

\# Summary Statistics
          ID  group   age     sex    BMI  patient\_contact
              using\_FFP2\_mask  negative\_swab  booster
              positive\_household  months\_since\_immunisation
               time\_dose1\_to\_dose\_2      vaccinetype  day\
              _interval\_start  day\_interval\_stop  infection\
              _event
```

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```
count  12086  12086 12065   11987  12086            11686
              11686          12086    12086              12086
                          11459                   9332
              10035               12086              12086
              12086
unique   NaN      4   NaN        2       2              NaN
                NaN            NaN       NaN                2
                            NaN                    NaN
                  3                 NaN                NaN
                NaN
top      NaN      V   NaN  female    u30              NaN
                NaN            NaN       NaN                 no
                            NaN                    NaN  Pfizer\
   _BioNTech                 NaN                 NaN
                NaN
freq     NaN   8157   NaN    9617  10625              NaN
                NaN            NaN       NaN              10584
                            NaN                    NaN
               7816                 NaN                NaN
                NaN
mean    1300    NaN 44.03     NaN     NaN           0.7941
             0.2014         0.4933   0.5007                NaN
                          5.015                  1.026
                NaN                81.21              113.2
            0.06321
std    748.2    NaN 11.01     NaN     NaN           0.4044
             0.4011            0.5       0.5                NaN
                          2.344                  0.4213
                NaN                47.03               32.1
             0.2434
min        1    NaN    17     NaN     NaN                0
                0              0         0                NaN
                          -5.3                       0
                NaN                     0                  1
                  0
25\%     648    NaN    35     NaN     NaN                1
                  0              0         0                NaN
                            3.8                      0.9
                NaN                    75                  88
                  0
50\%    1310    NaN    44     NaN     NaN                1
                  0              0         1                NaN
                            5.5                        1
                NaN                    99                 106
                  0
75\%    1942    NaN    53     NaN     NaN                1
                  0              1         1                NaN
                            6.6                      1.2
                NaN                   113                 142
```

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```
                    0
max      2595     NaN     73      NaN     NaN                  1
                    1                1         1                  NaN
                             17.8                     5.1
                  NaN                   171                  178
                    1

           ID group    age      sex   BMI   comorbidity   using\_FFP2
               \_mask   months\_until\_reinfection   variant
               booster\_over7\_days\_before   symptom\_number
count     764    764    764      759   764           719
                  734                          764          764
                             764               764
unique    NaN      4    NaN        2     2           NaN
                  NaN                          NaN          2
                             NaN               NaN
top       NaN      V    NaN   female   u30           NaN
                  NaN                          NaN   omicron
                             NaN               NaN
freq      NaN    550    NaN      620   679           NaN
                  NaN                          NaN       591
                             NaN               NaN
mean     1315    NaN  41.45      NaN   NaN        0.3825
    0.1839                        4.1       NaN
                        0.5209                3.806
std     742.7    NaN  10.69      NaN   NaN        0.4863
    0.3877                       1.268       NaN
                        0.4999                2.177
min         2    NaN     17      NaN   NaN             0
                    0                            0       NaN
                               0                 0
25\%     694.8   NaN     33      NaN   NaN             0
                    0                          3.4       NaN
                               0                 2
50\%      1354   NaN     41      NaN   NaN             0
                    0                          4.4       NaN
                               1                 4
75\%      1942   NaN  49.25      NaN   NaN             1
                    0                          5.2       NaN
                               1                 5
max      2588    NaN     68      NaN   NaN             1
                    1                          5.8       NaN
                               1                11

\# Categorical Variables (Top 5 categories including
    unvaccinated represented as '')
group: group
V     8157
H     2996
```

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```
I     469
N     464
sex: sex
female    9617
male      2370
NaN         99
BMI: BMI
u30    10625
o30     1461
positive\_household: positive\_household
no     10584
yes     1502
vaccinetype: vaccinetype
Pfizer\_BioNTech    7816
Moderna            2201
NaN                2051
other\_unknown        18
group: group
V    550
H    110
N     67
I     37
sex: sex
female    620
male      139
NaN         5
BMI: BMI
u30    679
o30     85
variant: variant
omicron    591
delta      173

\# Missing Values
TimeToInfection.csv: ID                                0
group                              0
age                               21
sex                               99
BMI                                0
patient\_contact                 400
using\_FFP2\_mask                 400
negative\_swab                     0
booster                            0
positive\_household                0
months\_since\_immunisation      627
time\_dose1\_to\_dose\_2          2754
vaccinetype                    2051
day\_interval\_start                0
day\_interval\_stop                 0
```

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```
infection\_event                 0

Symptoms.csv: ID                            0
group                          0
age                            0
sex                            5
BMI                            0
comorbidity                   45
using\_FFP2\_mask               30
months\_until\_reinfection       0
variant                        0
booster\_over7\_days\_before      0
symptom\_number                 0
\# Interval Lengths
Interval length statistics: count   12086
mean    32.01
std     26.43
min         1
25\%        14
50\%        24
75\%        34
max        99
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
import pandas as pd
import numpy as np
import pickle
from statsmodels.discrete.discrete_model import Logit
from statsmodels.regression.linear_model import OLS

# LOAD DATA
df_time_to_infection = pd.read_csv('TimeToInfection.csv')

# DATASET PREPARATIONS
# Calculate the duration for each interval
df_time_to_infection['interval_length'] = df_time_to_infection[
    ↪ 'day_interval_stop'] - df_time_to_infection['
    ↪ day_interval_start']

# Aggregate data by ID to get total time at risk and whether an
    ↪  infection occured
```

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```
df_time_to_infection = df_time_to_infection.groupby('ID').agg({
    ↪ 'interval_length':'sum', 'group':'first',
  'age':'first', 'sex':'first', 'BMI': 'first',
  'infection_event':'max'}).reset_index()
df_time_to_infection.rename(columns={'interval_length': '
    ↪ total_time_at_risk'}, inplace=True)

# DESCRIPTIVE STATISTICS
# No descriptive statistics table is needed.

# PREPROCESSING
# Handle missing data
df_time_to_infection.loc[df_time_to_infection['sex'] == '', '
    ↪ sex'] = np.NaN
df_time_to_infection.loc[df_time_to_infection['age'].isna(), '
    ↪ age'] = df_time_to_infection['age'].mean()

# Create dummy variables for the 'group' column
df = pd.get_dummies(df_time_to_infection, columns=['group'],
    ↪ drop_first=False)

# ANALYSIS
## Table 1: "Impact of vaccination group on infection event and
    ↪  total time at risk"
model = Logit.from_formula('infection_event ~
    ↪ total_time_at_risk + group_V + group_I + group_H', df)
result = model.fit()
df1 = pd.DataFrame(result.summary2().tables[1])
df1.to_pickle('table_1.pkl')

## Table 2: "Vaccination group impact on total time at risk
    ↪ among infected individuals"
df_infected = df[df['infection_event'] == 1]
# As the sum of group_V, group_I, group_H, and group_N equals
    ↪ 1, we can remove group_N to avoid singularity
model2 = OLS.from_formula('total_time_at_risk ~ group_V +
    ↪ group_I + group_H', df_infected)
result2 = model2.fit()
df2 = pd.DataFrame(result2.summary2().tables[1])
df2.to_pickle('table_2.pkl')

## Table 3: "Distribution of infection events by vaccination
    ↪ group"
grouped_df = df_time_to_infection.groupby('group').agg({'
    ↪ total_time_at_risk':'mean', 'infection_event':'sum'})
grouped_df['infection_rate'] = grouped_df['infection_event'] /
    ↪ grouped_df['total_time_at_risk']
df3 = grouped_df[['infection_rate']]
```

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```
df3.to_pickle('table_3.pkl')

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': df.shape[0],
    'Log-Likelihood of model': result.llf,
    'AIC of model with Total Time at Risk Among Infected
        ↪ Individuals': result2.aic
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Code Description

The provided code performs data analysis on the dataset "TimeToInfection.csv" to investigate the impact of vaccination status on the risk of infection among healthcare workers over time.

The code first calculates the total time at risk for each individual by aggregating the time intervals and identifying if an infection event occurred. It then preprocesses the data by handling missing values and creating dummy variables for the vaccination groups.

Three main analyses are conducted: 1. The code fits a logistic regression model to determine the impact of vaccination group (twice vaccinated, infected unvaccinated, hybrid) on the likelihood of infection while considering the total time at risk. 2. An OLS regression model is used to assess how vaccination status affects the total time at risk specifically among infected individuals. 3. The distribution of infection events by vaccination group is explored to understand the infection rates across different groups.

The results of these analyses are saved as pickled files "table_1.pkl", "table_2.pkl", and "table_3.pkl" respectively. Additionally, the code saves the total number of observations, the log-likelihood of the model, and the AIC value of the model focusing on total time at risk among infected individuals in the "additional_results.pkl" file for further examination.

C.3 Code Output

table_1.pkl

```
                    Coef.  Std.Err.        z      P$>$\textbar
                       {}z\textbar{}  [0.025    0.975]
Intercept           4.305     0.2915    14.77  2.23e-49    3.734
        4.877
```

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```
group\_V[T.True]       0.2428    0.2036   1.193     0.233
    -0.1562   0.6418
group\_I[T.True]      -0.0613    0.2781 -0.2204     0.826
    -0.6064   0.4838
group\_H[T.True]      -0.4777    0.2265  -2.109    0.0349
    -0.9216 -0.03378
total\_time\_at\_risk -0.03596  0.001752  -20.52  1.31e-93
    -0.0394 -0.03253
```

table_2.pkl

```
                  Coef.  Std.Err.     t     P$>$\textbar{}t\
                     textbar{}  [0.025  0.975]
Intercept         104.4     4.292 24.32  4.42e-97   95.95
    112.8
group\_V[T.True]  19.19     4.592  4.18  3.25e-05   10.18
    28.21
group\_I[T.True]   24.4     7.199 3.389  0.000738   10.26
    38.53
group\_H[T.True]  31.17     5.677  5.49  5.49e-08   20.02
    42.31
```

table_3.pkl

```
       infection\_rate
group
H               0.6697
I               0.2854
N               0.6358
V                3.607
```

additional_results.pkl

```
{
    'Total number of observations': 2595,
    'Log-Likelihood of model': -1242            ,
    'AIC of model with Total Time at Risk Among Infected
        Individuals': 7726              ,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```python
# IMPORT
import pandas as pd
```

Created by data-to-paper (AI)

```python
from my_utils import to_latex_with_note, is_str_in_df,
    split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES
shared_mapping: AbbrToNameDef = {
    'Coef.': ('Coefficient', None),
    'Std.Err.': ('Std. Error', None),
    'z': ('z-value', None),
    'P>|z|':('p-value', None),
    't': ('t-value', None),
    'P>|t|':('p-value', None),
    'BMI': ('BMI Category','"o30" for over 30  or "u30" for
        below 30'),
}

# TABLE 1
df1 = pd.read_pickle('table_1.pkl')

## RENAME ROWS AND COLUMNS
mapping1 = dict((k, v) for k, v in shared_mapping.items() if
    is_str_in_df(df1, k))
mapping1 |= {
    'group_V[T.True]': ('Vaccinated Group', 'Twice vaccinated,
        uninfected'),
    'group_I[T.True]': ('Infected Group', 'Infected,
        unvaccinated'),
    'group_H[T.True]': ('Hybrid Group', 'Infected and $\geq$1
        vaccination, referred to as Hybrid immunity'),
    'total_time_at_risk': ('Total Time at Risk', 'Total time at
        risk for infection in the given group')
}
abbrs_to_names1, legend1 = split_mapping(mapping1)
df1 = df1.rename(columns=abbrs_to_names1, index=abbrs_to_names1
    )

# SAVE AS LATEX
to_latex_with_note(
    df1, 'table_1.tex',
    caption="Impact of vaccination group on infection event and
        total time at risk",
    label='table:table1EffectOfVaccinationGroup',
    legend=legend1)

# TABLE 2
df2 = pd.read_pickle('table_2.pkl')

## RENAME ROWS AND COLUMNS
mapping2 = dict((k, v) for k, v in shared_mapping.items() if
    is_str_in_df(df2, k))
```

Created by data-to-paper (AI)

```
mapping2 |= {
    'group_V[T.True]': ('Vaccinated Group', 'Twice vaccinated,
        ↪ uninfected'),
    'group_I[T.True]': ('Infected Group', 'Infected,
        ↪ unvaccinated'),
    'group_H[T.True]': ('Hybrid Group', 'Infected and $\geq$1
        ↪ vaccination, referred to as Hybrid immunity')
}
abbrs_to_names2, legend2 = split_mapping(mapping2)
df2 = df2.rename(columns=abbrs_to_names2, index=abbrs_to_names2
    ↪ )

# SAVE AS LATEX
to_latex_with_note(
    df2, 'table_2.tex',
    caption="Vaccination group impact on total time at risk
        ↪ among infected individuals",
    label='table:table2VaccinatedGroupTimeAtRisk',
    legend=legend2)

# TABLE 3
df3 = pd.read_pickle('table_3.pkl')

## RENAME ROWS AND COLUMNS
mapping3 = dict((k, v) for k, v in shared_mapping.items() if
    ↪ is_str_in_df(df3, k))
mapping3 |= {
    'H': ('Hybrid Immunity Group', 'Infected and $\geq$1
        ↪ vaccination, referred to as Hybrid immunity'),
    'I': ('Infected Group', 'Infected, unvaccinated'),
    'N': ('No Immunity Group', 'No immunity'),
    'V': ('Vaccinated Group', 'Twice vaccinated, uninfected'),
    'infection_rate': ('Infection Rate', 'Rate of infection
        ↪ within the respective group')
}
abbrs_to_names3, legend3 = split_mapping(mapping3)
df3 = df3.rename(columns=abbrs_to_names3, index=abbrs_to_names3
    ↪ )

# SAVE AS LATEX
to_latex_with_note(
    df3, 'table_3.tex',
    caption="Distribution of infection events by vaccination
        ↪ group",
    label='table:table3InfectionDistribution',
    legend=legend3)
```

Created by data-to-paper (AI)

D.2 Provided Code

The code above is using the following provided functions:

```
def to_latex_with_note(df, filename: str, caption: str, label:
    ↪ str, note: str = None, legend: Dict[str, str] = None, **
    ↪ kwargs):
    """
    Converts a DataFrame to a LaTeX table with optional note
        ↪ and legend added below the table.

    Parameters:
    - df, filename, caption, label: as in `df.to_latex`.
    - note (optional): Additional note below the table.
    - legend (optional): Dictionary mapping abbreviations to
        ↪ full names.
    - **kwargs: Additional arguments for `df.to_latex`.
    """

def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, '
        ↪ levels', [df.index]) + getattr(df.columns, 'levels',
        ↪ [df.columns]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef
    ↪ ):
    abbrs_to_names = {abbr: name for abbr, (name, definition)
        ↪ in abbrs_to_names_and_definitions.items() if name is
        ↪ not None}
    names_to_definitions = {name or abbr: definition for abbr,
        ↪ (name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

table_1.tex

```
\% This latex table was generated from: `table\_1.pkl`
\begin{table}[h]
\caption{Impact of vaccination group on infection event and
    total time at risk}
\label{table:table1EffectOfVaccinationGroup}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{\%
```

Created by data-to-paper (AI)

```
\begin{tabular}{lllllll}
\toprule
 \& Coefficient \& Std. Error \& z-value \& p-value \& [0.025
     \& 0.975] \\
\midrule
\textbf{Intercept} \& 4.31 \& 0.291 \& 14.8 \& \$$<$\$1e-06 \&
    3.73 \& 4.88 \\
\textbf{Vaccinated Group} \& 0.243 \& 0.204 \& 1.19 \& 0.233 \&
     -0.156 \& 0.642 \\
\textbf{Infected Group} \& -0.0613 \& 0.278 \& -0.22 \& 0.826
    \& -0.606 \& 0.484 \\
\textbf{Hybrid Group} \& -0.478 \& 0.226 \& -2.11 \& 0.0349 \&
    -0.922 \& -0.0338 \\
\textbf{Total Time at Risk} \& -0.036 \& 0.00175 \& -20.5 \& \
    $$<$\$1e-06 \& -0.0394 \& -0.0325 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Vaccinated Group}: Twice vaccinated, uninfected
\item \textbf{Infected Group}: Infected, unvaccinated
\item \textbf{Hybrid Group}: Infected and \$\geq\$1 vaccination
    , referred to as Hybrid immunity
\item \textbf{Total Time at Risk}: Total time at risk for
    infection in the given group
\end{tablenotes}
\end{threeparttable}
\end{table}
```

table_2.tex

```
\% This latex table was generated from: `table\_2.pkl`
\begin{table}[h]
\caption{Vaccination group impact on total time at risk among
    infected individuals}
\label{table:table2VaccinatedGroupTimeAtRisk}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{\%
\begin{tabular}{lrrrlrr}
\toprule
 \& Coefficient \& Std. Error \& t-value \& p-value \& [0.025
     \& 0.975] \\
\midrule
\textbf{Intercept} \& 104 \& 4.29 \& 24.3 \& \$$<$\$1e-06 \&
    95.9 \& 113 \\
\textbf{Vaccinated Group} \& 19.2 \& 4.59 \& 4.18 \& 3.25e-05
    \& 10.2 \& 28.2 \\
\textbf{Infected Group} \& 24.4 \& 7.2 \& 3.39 \& 0.000738 \&
```

Created by data-to-paper (AI)

```
    10.3 \& 38.5 \\
\textbf{Hybrid Group} \& 31.2 \& 5.68 \& 5.49 \& \$$<$\$1e-06
    \& 20 \& 42.3 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Vaccinated Group}: Twice vaccinated, uninfected
\item \textbf{Infected Group}: Infected, unvaccinated
\item \textbf{Hybrid Group}: Infected and \$\geq\$1 vaccination
    , referred to as Hybrid immunity
\end{tablenotes}
\end{threeparttable}
\end{table}
```

table_3.tex

```
\% This latex table was generated from: `table\_3.pkl`
\begin{table}[h]
\caption{Distribution of infection events by vaccination group}
\label{table:table3InfectionDistribution}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{\%
\begin{tabular}{lr}
\toprule
 \& Infection Rate \\
group \&  \\
\midrule
\textbf{Hybrid Immunity Group} \& 0.67 \\
\textbf{Infected Group} \& 0.285 \\
\textbf{No Immunity Group} \& 0.636 \\
\textbf{Vaccinated Group} \& 3.61 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Hybrid Immunity Group}: Infected and \$\geq\$1
    vaccination, referred to as Hybrid immunity
\item \textbf{Infected Group}: Infected, unvaccinated
\item \textbf{No Immunity Group}: No immunity
\item \textbf{Vaccinated Group}: Twice vaccinated, uninfected
\item \textbf{Infection Rate}: Rate of infection within the
    respective group
\end{tablenotes}
\end{threeparttable}
\end{table}
```

Created by data-to-paper (AI)